

2026 IEEE SUSTAINABLE POWER AND ENERGY CONFERENCE

iSPEC 2026

December 4–6, 2026 | Mar Baselios College of Engineering and Technology, Thiruvananthapuram, Kerala, India



SPECIAL SESSION 1 (SS1) • CP-SRRPS

Cyber-Physical Security, Reliability & Resiliency in Smart Power Systems

— Call for Papers —

Organized and Co-Chaired by:

- **Dr. Rohit K. Mathew** — Associate Professor, Dept. of EEE, The National Institute of Engineering, Mysuru
- **Dr. Vinu Thomas** — Associate Professor, École Centrale de Nantes (ECN), France
- **Dr. Nimal Madhu M.** — Assistant Professor, Dept. of Electrical Engineering, NIT Calicut

Session Overview:

The rapid integration of renewable energy resources, power electronic converters, intelligent control systems, and communication networks is transforming power systems into highly interconnected cyber-physical infrastructures — introducing vulnerabilities such as cyberattacks, communication failures, and false-data injection. This session provides a multidisciplinary forum to examine cyber-physical threats to smart grids, renewable-energy systems, and power-electronics-dominated networks, and to explore resilient protection, control, and AI-enabled detection strategies, supporting iSPEC 2026's focus on Grid Infrastructure & Resilience and Smart Grids, AI & Digitalization.

Topics of Interest Include, but Are Not Limited to:

- Cyber-physical security of smart grids and modern power systems
- Cyber-resilient control of power electronic converters and inverter-based resources
- Security and resilience of microgrids and networked microgrids
- AI-enabled cyber-physical situational awareness
- Cyber-resilient state estimation and system identification
- Resilience assessment, metrics, and enhancement of smart power systems
- Resilient operation under cyberattacks, extreme events, and cascading failures
- Digital twins for power-system security, reliability, and resilience
- Hardware-in-the-loop and real-time simulation for cyber-physical security
- Reliability and resilience of converter-dominated, low-inertia power systems

Important Dates:

Paper Submission Opens	25 August 2026
Paper Submission Deadline	30 September 2026
Acceptance Notification	15 October 2026
Registration Due	1 November 2026
Conference Dates	4–6 December 2026

Format: Half-day technical special session, targeting ~8–10 peer-reviewed papers. Author guidelines and peer-review standards will follow the regular iSPEC 2026 paper submission process. Accepted and presented papers will be submitted for inclusion in IEEE Xplore, subject to IEEE Xplore's scope and quality requirements.

Submission Portal: <https://ispec2026.org/submission>

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